

AI SDLC Copilot — Multi-Agent Architecture Blueprint

1. Simple Overview (Flow)

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graph TD;
    A[User Upload] --> B[Document Ingestion Layer];
    B --> C[Preprocessing & Context Layer];
    C --> D[Planner / Orchestrator Agent];
    D --> E[Requirement Understanding Layer];
    E --> F[Validation & Risk Detection Layer];
    F --> G[Artifact Generation Layer];
    G --> H[Architecture & Governance Layer];
    H --> I[Quality Validation Layer];
    I --> J[Human Feedback Layer];
    J --> K[Sprint Planning & Resource Allocation Layer];
    K --> L[Export & Delivery Layer];
    L --> M[Dashboard / Jira / GitHub];
```

2. Workflow and Agents in Order

1. Document Ingestion Agent
2. Text Cleaning Agent
3. Chunking & Context Agent
4. Planner / Orchestrator Agent
5. Requirement Extraction Agent
6. Requirement Classification Agent

7. Ambiguity Detection Agent
8. Missing Requirement Agent
9. Conflict Detection Agent
10. User Story Generation Agent
11. Task Decomposition Agent
12. Acceptance Criteria Agent
13. Test Case Generation Agent
14. API Specification Agent
15. Database Schema Agent
16. Security Review Agent
17. Scalability & Architecture Agent
18. Effort Estimation Agent
19. Hallucination Validation Agent
20. Traceability Agent
21. Feedback & Refinement Agent
22. Sprint Planning Agent
23. Team Allocation & Jira Assignment Agent
24. Export & Integration Agent
25. Dependency Mapping Agent
26. DevOps Recommendation Agent
27. Compliance Agent

3. Agents, Responsibilities, Purpose & Edge Cases

1. Document Ingestion Agent

Purpose: Accept requirement inputs into SDLC system. Responsibilities: OCR, PDF parsing, metadata extraction, format normalization. Edge Cases: Corrupted PDFs, empty files, scanned images, unsupported formats, multi-language docs.

2. Text Cleaning Agent

Purpose: Clean noisy requirement data. Responsibilities: Remove duplicates, normalize text, remove headers/footers. Edge Cases: OCR garbage, HTML artifacts, broken formatting, inconsistent bullets.

3. Chunking & Context Agent

Purpose: Prepare requirements for LLM processing. Responsibilities: Semantic chunking, embedding generation, context preservation. Edge Cases: Token overflow, duplicate chunks, broken context, cross-page references.

4. Planner / Orchestrator Agent

Purpose: Route workflow between agents. Responsibilities: Agent selection, execution management, retries, output aggregation. Edge Cases: Agent failures, infinite loops, timeout handling, contradictory outputs.

5. Requirement Extraction Agent

Purpose: Convert raw text into structured engineering entities. Responsibilities: Extract actors, actions, modules, dependencies, workflows. Edge Cases: Ambiguous actors, nested workflows, implicit logic, poorly written requirements.

6. Requirement Classification Agent

Purpose: Categorize requirements. Responsibilities: Identify functional, non-functional, security, performance, compliance requirements. Edge Cases: Mixed categories, hidden NFRs, incorrect classifications.

7. Ambiguity Detection Agent

Purpose: Detect unclear requirements. Responsibilities: Identify vague terms, weak verbs, missing thresholds. Edge Cases: Subjective wording, undefined acronyms, conflicting interpretations.

8. Missing Requirement Agent

Purpose: Identify incomplete requirement details. Responsibilities: Detect missing validations, permissions, retry logic, workflows. Edge Cases: Missing edge flows, missing audit rules, missing error handling.

9. Conflict Detection Agent

Purpose: Detect contradictory requirements. Responsibilities: Identify duplicates, conflicting logic, inconsistent workflows. Edge Cases: Semantic conflicts, version mismatches, cross-document contradictions.

10. User Story Generation Agent

Purpose: Generate Agile stories. Responsibilities: Convert requirements into structured user stories. Edge Cases: Missing actors, multiple personas, complex user journeys.

11. Task Decomposition Agent

Purpose: Generate engineering tasks. Responsibilities: Create frontend, backend, API, DB, QA, DevOps tasks. Edge Cases: Circular dependencies, duplicate tasks, unrealistic decomposition.

12. Acceptance Criteria Agent

Purpose: Define validation rules. Responsibilities: Generate business conditions and success scenarios. Edge Cases: Missing negative conditions, incomplete flows, vague acceptance rules.

13. Test Case Generation Agent

Purpose: Generate QA artifacts. Responsibilities: Create positive, negative, boundary, regression, Gherkin tests. Edge Cases: Invalid inputs, permission failures, concurrency issues, timeout scenarios.

14. API Specification Agent

Purpose: Generate backend API contracts. Responsibilities: Create endpoints, request/response schemas, status codes. Edge Cases: Broken contracts, missing validation, wrong HTTP methods.

15. Database Schema Agent

Purpose: Generate storage architecture. Responsibilities: Create tables, relationships, indexes, constraints. Edge Cases: Missing foreign keys, normalization issues, scalability bottlenecks.

16. Security Review Agent

Purpose: Detect security risks. Responsibilities: Analyze auth, RBAC, encryption, uploads, secrets. Edge Cases: Plain-text passwords, token leakage, insecure uploads, injection risks.

17. Scalability & Architecture Agent

Purpose: Recommend scalable system design. Responsibilities: Suggest caching, queues, load balancing, microservices. Edge Cases: SPOF, traffic bottlenecks, missing cache strategy, scaling assumptions.

18. Effort Estimation Agent

Purpose: Estimate engineering effort. Responsibilities: Generate story points, timelines, sprint estimates. Edge Cases: Unrealistic deadlines, ignored dependencies, underestimated infra work.

19. Hallucination Validation Agent

Purpose: Validate AI-generated outputs. Responsibilities: Detect fake APIs, invalid logic, impossible architectures. Edge Cases: Fabricated assumptions, hallucinated dependencies, inconsistent outputs.

20. Traceability Agent

Purpose: Maintain SDLC traceability. Responsibilities: Map requirements → stories → tasks → APIs → tests. Edge Cases: Orphan tasks, missing test coverage, duplicate mappings.

21. Feedback & Refinement Agent

Purpose: Human-in-the-loop governance. Responsibilities: Accept edits, selective regeneration, approvals, rollback. Edge Cases: Conflicting edits, accidental overwrite, invalid refinements.

22. Sprint Planning Agent

Purpose: Convert tasks into sprint plans. Responsibilities: Sprint grouping, dependency sequencing, workload balancing. Edge Cases: Sprint overload, blocked tasks, dependency conflicts.

23. Team Allocation & Jira Assignment Agent

Purpose: Assign Jira tasks intelligently. Responsibilities: Map tasks to team members based on roles, availability, workload, sprint capacity, effort estimates. Edge Cases: Resource overload, unavailable engineers, skill mismatch, overlapping schedules, sprint changes.

24. Export & Integration Agent

Purpose: Deliver outputs to external systems. Responsibilities: Export to Jira, GitHub, CSV, PDF, Confluence. Edge Cases: Invalid payloads, API limits, schema mismatch, auth failures.

25. Dependency Mapping Agent

Purpose: Manage execution dependencies. Responsibilities: Build dependency graphs, detect blockers, identify critical paths. Edge Cases: Circular dependencies, hidden blockers, invalid ordering.

26. DevOps Recommendation Agent

Purpose: Suggest deployment workflows. Responsibilities: Generate CI/CD, Docker, monitoring, deployment recommendations. Edge Cases: Missing rollback, environment mismatch, deployment failures.

27. Compliance Agent

Purpose: Validate regulatory compliance. Responsibilities: Check GDPR, HIPAA, PCI-DSS, SOC2, PII handling. Edge Cases: Missing encryption, audit gaps, improper storage, compliance violations.

Recommended Stack

Backend: Python + FastAPI
Orchestration: LangGraph, CrewAI, AutoGen, PydanticAI
LLMs: GPT-4o, Claude, Gemini, Groq
OSS Models: spaCy, Hugging Face
Vector DB: Pinecone, ChromaDB, OpenSearch
Database: PostgreSQL, MongoDB
Deployment: Docker, AWS ECS/Fargate, Kubernetes